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Device and method for providing a bone cement paste

Abstract

A device for providing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow cylindrical cartridge with an interior space, wherein a discharge piston axially movable in the interior space is arranged in the space, which piston divides the interior space into a proximal part of the interior space and a distal part of the interior space, wherein the proximal part and the distal part of the interior space are connected to one another in a fluid-conducting manner via a conduit means, wherein a bone cement powder is stored in the proximal part of the interior space as the first starting component, and wherein a conveying piston axially movable in the interior space is arranged in the distal part of the interior space, and a reservoir for a monomer liquid as a second starting component, which is connected or connectable in a fluid-conducting manner.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority pursuant to 35 U.S.C. 119(a) to European Application No. 22185605.7, filed Jul. 19, 2022, which application is incorporated herein by reference in its entirety.
FIELD OF THE INVENTION
(2) The invention relates to a device for providing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow cylindrical cartridge with an interior space, wherein a discharge piston axially movable in the interior space is arranged in the interior space, which piston divides the interior space into a proximal part of the interior space and a distal part of the interior space, wherein the proximal part and the distal part of the interior space are connected to one another in a fluid-conducting manner via a conduit means, wherein a bone cement powder is stored in the proximal part of the interior space as the first starting component, and wherein a conveying piston axially movable in the interior space is arranged in the distal part of the interior space, and a reservoir for a monomer liquid as a second starting component, which is connected or connectable in a fluid-conducting manner via an inlet channel to the distal part of the interior space for introducing the monomer liquid from the reservoir into the mixing unit.
(3) The invention further relates to a method for providing a bone cement paste from two starting components by means of such a device.
BACKGROUND OF THE INVENTION
(4) Considerable efforts are being made to demonstrate devices and methods for providing bone cement by means of which bone cement paste can be provided easily, reliably, and quickly. One important aspect of providing bone cement paste is the avoidance of air inclusions, e.g., gas bubbles, in the bone cement. To avoid this, a plurality of vacuum cementing systems have been described, of which the following are mentioned by way of example: U.S. Pat. No. 6,033,105 A, 5,624,184 A, 4,671,263 A, 4,973,168 A, 5,100,241 A, WO 99/67015 A1, EP 1020167 A2, US 5,586,821 A, EP 1016452 A2, DE 3640279 A1, WO 94/26403 A1, EP 1005901 A2, EP 1886647 A1, and U.S. Pat. No. 5,344,232 A.
(5) Within the market, there is a desire to simplify the provision of bone cement paste. One development consists of developing cementing systems in which both starting components are stored in separate regions of the mixing systems and are only mixed with one another in the cementing system immediately before the cementing application. Such closed, so-called full-prepacked systems are mentioned, for example, in the following publications: EP 0 692 229 A1, DE 10 2009 031 178 B3, U.S. Pat. No. 5,997,544 A, 6,709,149 B1, DE 698 12 726 T2, EP 0 796 653 A2, and US 5,588,745 A.
(6) In the aforementioned full-prepacked systems, the mixing of a monomer liquid with a bone cement powder is carried out by mechanical mixing, for example by means of a mixing rod.
(7) In contrast to the aforementioned full-prepacked systems, the specification EP 3 320 870 B1 describes a device in which the mixing of a monomer liquid with a bone cement powder takes place merely by pressing the monomer liquid into, in particular compacted, bone cement powder. The device described therefore does not require mechanical mixing, in particular a mixing rod.

Consequently, such devices are designed without mixing devices.
(8) In the device, a container filled with monomer liquid is stored axially behind a region filled with a bone cement powder inside a cartridge. A discharge piston is arranged between the bone cement powder and the container. In order to provide a bone cement paste, a conveying piston, which is arranged on a side of the container opposite the discharge piston, is driven forward in the direction of the discharge piston, resulting in opening of the container, in particular by partial shattering of a container in the form of a glass ampoule into container parts. The monomer liquid exiting the container is conveyed into the bone cement powder by continued advancement of the conveying piston, forming the bone cement paste.
(9) Comparable devices are also described in the specifications EP 3 320 869 B1 and EP 3 403 716 B1.
(10) A disadvantage of these devices is that in order to open and convey the monomer liquid into the bone cement powder, the container must be substantially completely destroyed, which on the one hand requires a comparatively high effort on the part of a user of the device and on the other hand makes it difficult to convey the monomer liquid into the bone cement powder substantially completely, in particular because of the fragments of the container.
(11) The patent application EP3838391 A1 also describes a device without mixing apparatus for providing bone cement paste.
(12) In this device, a container filled with a monomer liquid is stored in a reservoir, from which the monomer liquid, after opening the container, can flow via a conduit means into a distal part of an interior space before being conveyable by propelling a piston into a bone cement powder in a proximal part of the interior space. In this case, the reservoir is arranged such that the container is neither opened by the piston nor compressed when the monomer liquid is conveyed into the bone cement powder.
(13) A disadvantage of this device is that the monomer liquid can only flow slowly and/or intermittently through the conduit means into the distal part of the interior space, since the diameter of the conduit means must be designed such that the container, or parts thereof, cannot pass through the conduit means into the interior space. Furthermore, it is only with difficulty that a gas displaced from the interior space by the monomer liquid flowing into the interior can be discharged through the conduit means. The conduit means is therefore not designed for good mass transfer between reservoir and interior space. This is a disadvantage, since a fast, safe and substantially complete provision of the monomer liquid for mixing the bone cement paste is required, especially for time-critical operations. A further disadvantage of the device is its rather complex design with many moving parts. In addition, the piston, which has already been used to convey the monomer liquid into the bone cement powder, is not designed to discharge the bone cement paste from the device. In particular, the reservoir protruding from the device would also make it difficult to discharge the bone cement paste from the device for steric reasons.
(14) There is therefore demand in the market for further simplification of devices for providing bone cement paste.
Objects
(15) It is an object of the present invention to at least partially overcome one or more of the disadvantages resulting from the prior art.
(16) The invention is especially based on the goal of providing a device which permits simple and application-safe opening of one or more ampoules, in particular glass ampoules, with a monomer liquid for the simple, rapid, and application-safe providing of a bone cement paste. In particular, the opening of the ampoule or ampoules should take place with as little effort as possible and with avoidance of additional separate tools. Furthermore, the opening of the ampoule with as few components as possible should be enabled. Furthermore, the monomer liquid should be available with as little loss and as quickly as possible for providing the bone cement paste. Conveying the monomer liquid into a bone cement powder to provide the bone cement paste should be feasible

with as little force as possible.
(17) The device is to provide the bone cement paste without mechanical mixing of the starting components. The device should be able to provide the bone cement without an externally applied vacuum. The device should be able to be operated with as few steps as possible in order to minimize sources of error by the user.
(18) A further object of the invention is to provide a method with which bone cement can be provided from two starting components, by means of which at least some of the objects already described are achieved at least in part.
Preferred Embodiments of the Invention
(19) The features of the independent claims contribute to at least partially fulfilling at least one of the aforementioned objects. The dependent claims provide preferred embodiments which contribute to at least partially fulfilling at least one of the objects.
(20) A first embodiment of the invention is a device for providing a bone cement paste from two starting components, comprising: a mixing unit comprising a hollow cylindrical cartridge with an interior space, wherein a discharge piston axially movable in the interior space is arranged in the interior space, which piston divides the interior space into a proximal part of the interior space and a distal part of the interior space, wherein the proximal part and the distal part of the interior space are connected to one another in a fluid-conducting manner via a conduit means, wherein a bone cement powder is stored in the proximal part of the interior space as the first starting component, and wherein a conveying piston axially movable in the interior space is arranged in the distal part of the interior space, and a reservoir for a monomer liquid as a second starting component, which is connected or connectable in a fluid-conducting manner via an inlet channel to the distal part of the interior space for introducing the monomer liquid from the reservoir into the mixing unit, characterized in that the reservoir and the mixing unit are connected or connectable in a fluid-conducting manner via an outlet channel, in particular an outlet channel disjoint from the inlet channel, via which a gas can be discharged from the interior space into the reservoir, in particular in order to improve, in particular facilitate and/or accelerate, introduction of the monomer liquid into the mixing unit.
(21) In one embodiment of the device, the reservoir comprises a reservoir container in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head is arranged, and the monomer liquid is stored in the ampoule, and comprises a cavity in the region of the ampoule head, wherein the cavity is fluid-conductively connected to the inlet channel and comprises a connection to the ampoule, wherein the ampoule head is at least in regions disposed in the connection, and wherein the reservoir container at least in portions comprises a deformable region such that tilting of the ampoule about a pivot point against the connection is enabled. This embodiment is a second embodiment of the invention, which is preferably dependent upon the first embodiment of the invention.
(22) In one embodiment of the device, the inlet channel has a smaller distance to the pivot point than the outlet channel. This embodiment is a third embodiment of the invention, which is preferably dependent upon the second embodiment of the invention.
(23) In one embodiment of the device, the outlet channel opens into the interior space proximal to the inlet channel. This embodiment is a fourth embodiment of the invention which is preferably dependent upon one of the preceding embodiments of the invention.
(24) In one embodiment of the device, the inlet channel is formed as a funnel at an inlet channel end opposite the mixing unit. This embodiment is a fifth embodiment of the invention, which is preferably dependent on one of the preceding embodiments of the invention.
(25) In one embodiment of the device, the outlet channel has a minimum outlet channel diameter which corresponds to at least half the minimum inlet channel diameter of the inlet channel. This embodiment is a sixth embodiment of the invention, which is preferably dependent on one of the preceding embodiments of the invention.

(26) In one embodiment of the device, the inlet channel and the outlet channel are each formed in at least two parts, such that the mixing unit and the reservoir are connected to one another in a fluid-conducting manner in a first channel position of the inlet channel and the outlet channel, and are separated from one another in terms of fluid conduction in a second channel position of the inlet channel and the outlet channel. This embodiment is a seventh embodiment of the invention, which is preferably dependent upon one of the preceding embodiments of the invention.
(27) In one embodiment of the device, the device comprises a closure element which closes or makes closable, in terms of fluid conduction, at least a part of the at least two-part inlet channel and of the at least two-part outlet channel that faces the mixing unit. This embodiment is an eighth embodiment of the invention which is preferably dependent on the seventh embodiment of the invention.
(28) In one embodiment of the device, the mixing unit and the reservoir are reversibly connected or connectable to one another via a first form closure, in particular in the region of the inlet channel and the outlet channel. This embodiment is a ninth embodiment of the invention, which is preferably dependent on the eighth embodiment of the invention.
(29) In one embodiment of the device, the closure element is a rotary valve through which the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit extends and which, in a first rotary valve position, leaves the inlet channel and the outlet channel in the first channel position and, by rotating to a second rotary valve position, moves the inlet channel and the outlet channel to the second channel position. This embodiment is a tenth embodiment of the invention, which is preferably dependent on the eighth or ninth embodiment of the invention.
(30) In one embodiment of the device, after the reservoir has been separated from the mixing unit by releasing the first form closure, the closure element can be brought into a closure position in order to close, in terms of fluid conduction, the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit. This embodiment is an eleventh embodiment of the invention, which is preferably dependent on the ninth embodiment of the invention.
(31) In one embodiment of the device, the closure element is a screw. This embodiment is a twelfth embodiment of the device, which is preferably dependent on the eleventh embodiment of the invention.
(32) In one embodiment of the device, the mixing unit and the reservoir are reversibly connected or connectable to one another via a second form closure. This embodiment is a thirteenth embodiment of the invention, which is preferably dependent upon the ninth through twelfth embodiments of the invention.
(33) A fourteenth embodiment of the invention is a method for providing a bone cement paste from two starting components by means of a device in accordance with one of the preceding embodiments of the invention, comprising the steps of: a. the monomer liquid flowing from the reservoir through the inlet channel into the distal part of the interior space while simultaneously a gas from the interior space is discharged through the outlet channel into the reservoir, b. conveying the monomer liquid from the distal part of the interior space through the conduit means into the proximal part of the interior space by means of advancing the conveying piston in the direction of the discharge piston.
(34) In one embodiment of the method by means of a device according to any of the eighth to thirteenth embodiments of the invention, before the monomer liquid is conveyed in step b., the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit is closed in terms of fluid conduction by the closure element. This embodiment is a fifteenth embodiment of the invention, which is preferably dependent upon the fourteenth embodiment of the invention.
(35) General
(36) In the present description, range specifications also include the values specified as limits. A specification of the type “in the range of X to Y” with respect to a variable A consequently means that A can assume the values X, Y and values between X and Y. Ranges delimited on one side of

the type “up to Y” for a variable A accordingly mean, as a value, Y and less than Y.
(37) Some of the described features are linked to the term “substantially.” The term “substantially” is to be understood as meaning that, under real conditions and manufacturing techniques, a mathematically exact interpretation of terms such as “superimposition,” “perpendicular,” “diameter,” or “parallelism” can never be given exactly, but only within certain manufacturing-related error tolerances. For example, “substantially perpendicular axes” enclose an angle of 85 degrees to 95 degrees relative to one another, and “substantially equal volumes” comprise a deviation of up to 5% by volume. An “apparatus consisting substantially of plastic material” comprises, for example, a plastics content of ≥95 to ≤100% by weight. A “substantially complete filling of a volume B” comprises, for example, a filling of ≥95 to ≤100% by volume of the total volume of B.
(38) The terms “proximal” and “distal” are used only to designate the spatially opposite ends of the device or other structural units of the device and do not permit any inference of orientation with respect to a human body, such as a user of the device. “Distally to . . . ” and “proximally to . . . ” or similar formulations correspondingly express only the spatial arrangement of two structural units of the device in relation to one another.
DETAILED DESCRIPTION
(39) An initial subject of the invention relates to a device for providing a bone cement paste from two starting components, comprising: a mixing unit comprising a hollow cylindrical cartridge with an interior space, wherein a discharge piston axially movable in the interior space is arranged in the interior space, which piston divides the interior space into a proximal part of the interior space and a distal part of the interior space, wherein the proximal part and the distal part of the interior space are connected to one another in a fluid-conducting manner via a conduit means, wherein a bone cement powder is stored in the proximal part of the interior space as the first starting component, and wherein a conveying piston axially movable in the interior space is arranged in the distal part of the interior space, and a reservoir for a monomer liquid as a second starting component, which is connected or connectable in a fluid-conducting manner via an inlet channel to the distal part of the interior space for introducing the monomer liquid from the reservoir into the mixing unit, characterized in that the reservoir and the mixing unit are connected or connectable in a fluid-conducting manner via an outlet channel, via which a gas can be discharged from the interior space into the reservoir.
(40) The device is for mixing a bone cement paste of a bone cement powder and a monomer liquid, wherein prior to mixing the bone cement powder is stored in a mixing unit of the device and the monomer liquid can be stored in a reservoir of the device. Preferably, the reservoir stores at least one ampoule, preferably glass ampoule, filled with the monomer liquid. For example, the reservoir stores one or two ampoules, preferably one or two glass ampoules.
(41) The mixing unit is used for mixing the bone cement paste from the bone cement powder and the monomer liquid after conveying the monomer liquid into the mixing unit, in particular after conveying the monomer liquid into an interior space of the mixing unit.
(42) The mixing unit comprises a hollow cylindrical cartridge. A hollow cylindrical cartridge is to be understood as a tubular receptacle which comprises an interior space and a cartridge wall surrounding the interior space. The cross-section of the cartridge can assume any shape. Because the device is easy to manufacture and safe to use, the cross-section, and preferably also the cross-section of the interior space, is of circular design. This allows easy handling for the user and reduces a risk of movable parts wedging within the device, due to the absence of edges. According to the invention, the cartridge can consist of a wide variety of materials or material combinations. For example, the device can consist of a polymer. The polymer is preferably a transparent polymer, since this allows the user to visually check proper functioning of the device during use.
(43) A discharge piston is arranged in the interior space of the cartridges, which is axially movable in the interior space and divides the interior space into a proximal part of the interior space and a

distal part of the interior space. The bone cement powder is stored in the proximal part of the interior space, i.e., proximal to the discharge piston. Preferably, the discharge piston is equipped such that and cooperates with the cartridge wall such that substantially the bone cement powder cannot get into the distal part of the interior space.

(44) The discharge piston is further used to discharge the provided bone cement paste from the mixing unit. For this purpose, the discharge piston can be moved from its original position in the direction of a discharge opening of the mixing unit. The discharge opening is preferably located on a side of the bone cement powder axially opposite the discharge piston and thus proximal to the discharge piston. In order to remove a gas from the mixing unit, in particular the proximal part of the interior space, in particular before the bone cement paste is formed, it is preferred that the discharge opening is designed to be gas-permeable. For example, the discharge opening may be closed with a gas-conducting closure, such as a plug, which is removable from the discharge opening for discharging the mixed bone cement paste.

(45) The mixing unit comprises a conveying piston that can move axially in the interior space. The conveying piston is located in the distal part of the interior space, i.e., distal to the discharge piston. The conveying piston closes the mixing unit at a distal cartridge end in such that the monomer liquid conveyed from the reservoir into the distal part of the interior space cannot flow out of the cartridge. In an initial position of the device, the conveying piston is arranged in the interior space such that after the monomer liquid has been conveyed, it is stored between the discharge piston and the conveying piston in the distal part of the interior space.

(46) By advancing the conveying piston in the direction of the discharge piston, i.e., advancing in the proximal direction, the monomer liquid stored in the distal part of the interior space can be conveyed into the bone cement powder through a conduit means in the proximal part of the interior space, which connects the proximal part and the distal part of the interior space in a fluid-conducting manner. Fluid-conducting means that the distal part and the proximal part of the interior space are connected in a manner permeable to liquids—in particular, the monomer liquid—and to gases. In order to prevent bone cement powder from the proximal part from entering the distal part of the interior space, the conduit means is preferably equipped with a filter means, in particular a pore disc, for example made of sintered polypropylene particles, of sintered or compressed polyethylene fibers, of cellulose felt or of cardboard, which makes the conduit means impermeable to solids. In one variant of the device, at least one passage is provided in the discharge piston and/or between the discharge piston and the cartridge wall as conduit means through which the distal part and the proximal part of the interior space are connected to one another in a fluid-conducting manner. In this regard, a filter that is impermeable to the bone cement powder and permeable to the monomer liquid and gases, such as a pore disc made of, for example, sintered polypropylene particles, sintered or compressed polyethylene fibers, cellulose felt, or cardboard, may be disposed in or at one or both ends of the at least one passage. In a further variation of the device, the conduit means is one or more conduits arranged on the exterior of the cartridge or in the cartridge wall and connecting the distal part and the proximal part of the interior space. The discharge piston is bypassed in this variant.

(47) One variant of the device is designed such that continued advancement of the conveying piston in the direction of the discharge piston after the monomer liquid has been conveyed from the distal part of the interior space into the proximal part of the interior space causes the discharge piston to be advanced in the direction of the discharge opening of the device. In this manner, the bone cement paste provided by mixing bone cement powder and monomer liquid can be discharged from the device through the discharge opening. This is a simple manner of ensuring that the bone cement paste is expelled from the cartridge with the same drive as is used for conveying the monomer liquid, namely with the unidirectionally driven conveying piston.

(48) In order to prevent unintentional advancement of the discharge piston in the direction of the discharge opening, a detent means can be arranged on the discharge piston such that the discharge

piston can detent with the cartridge, in particular with the cartridge wall, wherein this detent cannot be released by the pressure to be applied when conveying the monomer liquid into the proximal part of the interior space, but can be released by a direct pressure of the conveying piston acting on the discharge piston.
(49) The detent means ensures that the monomer liquid can first be pressed into the bone cement powder, wherein the discharge piston maintains its original position relative to the cartridge and the interior space. Only after the monomer liquid has been largely pressed into the bone cement powder, and thus the bone cement paste is present in the proximal part of the interior space of the cartridge, can the bone cement paste then be pressed out of the proximal part of the cartridge with the discharge piston. Thus, the force required to release the detent is greater than the force required to convey the monomer liquid through the conduit means into the proximal part of the interior space.
(50) The reservoir is used to provide the monomer liquid before it provides the bone cement paste by mixing with the bone cement powder in the mixing unit. Preferably, the monomer liquid is stored in the reservoir until a user of the device wishes to provide the bone cement.
(51) To convey the monomer liquid from the reservoir to the distal part of the interior space of the mixing unit, the reservoir and the distal part of the interior space of the mixing unit are connected or connectable to one another via an inlet channel in a fluid-conducting manner. For this purpose, the inlet channel has an inlet channel diameter which allows the monomer liquid to be conveyed from the reservoir into the mixing unit as quickly as possible. For example, the inlet channel diameter, in particular a minimum inlet channel diameter, is in a range of 1 mm to 4 mm.
(52) In order to improve, in particular accelerate, the conveyance of the monomer liquid from the reservoir into the distal part of the interior space, the reservoir and the distal part of the interior space are, in addition to the inlet channel, connected or connectable to one another via an outlet channel, in particular one that is disjoint from the inlet channel, in a fluid-conducting manner.
(53) If the reservoir and the distal part of the interior space are connected to one another in a fluid-conducting manner via the inlet channel and the outlet channel, this permits improved, in particular accelerated, conveyance of the monomer liquid from the reservoir via the inlet channel into the distal part of the interior space of the mixing unit, while at the same time a gas from the distal part of the interior space, which is displaced from the distal part of the interior space by the monomer liquid entering the distal part of the interior space, can be discharged into the reservoir via the outlet channel. The inlet channel and the outlet channel thus synergistically ensure improved mass transfer between the reservoir and the mixing unit.
(54) Conveying of the monomer liquid from the reservoir via the inlet channel into the mixing unit can be triggered, for example, by gravity, by negative pressure in the mixing unit, in particular in the interior space of the cartridge, or a combination thereof, wherein conveying by means of gravity is preferred. Particularly when conveying by means of gravity, the outlet channel improves the introduction of the monomer liquid into the mixing unit, since the gas displaced from the interior space can be discharged into the reservoir.
(55) The reservoir can consist of a wide range of materials or material combinations. Examples, the reservoir may consist of a polymer. Preferably, the polymer is a transparent polymer, as this allows the user to visually monitor proper operation of the reservoir, particularly leakage of monomer liquid from the reservoir, during a use.
(56) The reservoir can be designed differently to provide the monomer liquid. For example, the monomer liquid may comprise a reservoir interior in which the monomer liquid is stored in a free-flowing manner. Preferably, the monomer liquid is stored within the reservoir in one or more separate receptacles, which facilitates handling and filling of the device, in particular the reservoir, and sterile provision of the monomer liquid. For example, the monomer liquid in the reservoir is provided in a container in the form of a bag. A bag is understood to be a non-rigid, largely flexible storage device capable of storing the monomer liquid in a hermetically sealed and sterile manner

and capable of being opened by means of the impact of an opening means, for example by piercing, cutting or tearing. The bags can be manufactured, for example, from a multilayer composite film—preferably comprising an EVOH barrier layer. Optionally, the bags can comprise a metal coating, and in particular an aluminum coating.

(57) Preferably, the reservoir stores at least one ampoule, preferably glass ampoule, containing the monomer liquid. For example, two ampoules containing the monomer liquid, preferably glass ampoules, are stored in the reservoir. Ampoules, in particular glass ampoules, are preferred due to good sterilizability and easy and reliable openability by manual application of force.

(58) An embodiment of the device is characterized in that the reservoir comprises a reservoir container in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head is arranged, and the monomer liquid is stored in the ampoule, and comprises a cavity in the region of the ampoule head, wherein the cavity is connected to the inlet channel in a fluid-conducting manner and comprises a connection to the ampoule, wherein the ampoule head is arranged at least in regions in the connection and the reservoir container comprises, at least in portions, a deformable region such that tilting of the ampoule about a pivot point against the connection is enabled.

(59) In this embodiment, the reservoir comprises a reservoir container for receiving one or more, preferably two, ampoules closed in terms of fluid conduction and filled with the monomer liquid, in particular a glass ampoule or glass ampoules, having an ampoule head and an ampoule body. The reservoir container surrounds the at least one ampoule, in particular at least the ampoule body, so that the ampoule can be stored securely in the reservoir until it is used. The reservoir container can, for example, be present in the form of a hollow cylinder into which the at least one ampoule is inserted, wherein, for improved transport capability of the device, the reservoir container is shaped such, for example by the reservoir container comprising a cover, that the ampoule cannot escape unintentionally from the device, in particular from the reservoir. The reservoir container is preferably shaped such that two ampoules, in particular two ampoules next to one another, can be stored, preferably with substantially parallel longitudinal axes, in the reservoir.

(60) Furthermore, the reservoir serves to open, in terms of fluid conduction, the at least one ampoule. For this purpose, the reservoir comprises a cavity which is connected via a connection to the ampoule arranged in the reservoir container. The ampoule is stored in the reservoir in such a way that the ampoule head points in the direction of the cavity, whereas the ampoule body is arranged at least partially, preferably entirely, in the reservoir container. The connection extends between cavity and reservoir container, and in fact in such a way that the ampoule head is arranged at least in regions in the connection. For this purpose, the connection has a connection diameter which allows the ampoule head to be inserted into the connection at least in portions. In one embodiment, the connection has a connection diameter which allows the ampoule head to be inserted completely into the connection. The connection diameter is preferably smaller than the diameter of the ampoule body so that the latter cannot be inserted into the connection. For example, the connection is designed as a ring or hollow cylinder, and the ampoule head is surrounded at least in regions by this ring or hollow cylinder. The connection has a structural integrity which exceeds a structural integrity of the ampoule, so that the ampoule can break when it is pressed against the connection.

(61) In order to open the ampoule or, given the presence of two or more ampoules, all ampoules, the reservoir container comprises a deformable region at least in portions, in particular adjacent to a transition of the ampoule head to the ampoule body of the ampoule. In one embodiment, the reservoir container is completely deformable. The deformable region allows tilting of the ampoule about a pivot point against the connection. The connection diameter is in this case matched to the ampoule head in such a way that, during tilting, at least the ampoule body end facing away from the ampoule head is tilted about the pivot point while at least the ampoule head end facing away from the ampoule body remains within the connection, so that the ampoule is opened in a fluid-

conducting manner by at least partial bursting of the ampoule, in particular in the region of an ampoule neck between ampoule head and ampoule body. The connection in this case serves primarily to fix the ampoule head against a tilting movement of the ampoule about the pivot point. For example, the connection diameter is not more than 10% larger than the diameter of the ampoule head so that a relatively slight tilting of the ampoule already leads to its opening in terms of fluid conduction.

(62) After opening of the at least one ampoule in terms of fluid conduction, the monomer liquid can flow out of the ampoule into the cavity. The cavity is connected to the mixing unit, in particular to the distal part of the interior space of the mixing unit, via the inlet channel in a fluid-conducting manner. Conveying of the monomer liquid from the cavity via the inlet channel into the mixing unit can be triggered, for example, by gravity, by a vacuum in the mixing unit, in particular in the interior space of the cartridge, or a combination thereof, wherein conveying by means of gravity is preferred. Particularly when conveying by means of gravity, the outlet channel improves the introduction of the monomer liquid into the mixing unit.

(63) An embodiment of the device is characterized in that the inlet channel has a smaller distance to the pivot point than the outlet channel. In particular, an inlet channel end opposite the mixing unit and facing the reservoir has a smaller distance, in particular a smaller spatial distance, to the pivot point than an outlet channel end opposite the mixing unit and facing the reservoir.

(64) The at least one ampoule is opened in a fluid-conducting manner when tilted about the pivot point against the connection in the vicinity of the pivot point, in particular in the region of the neck of the ampoule, such that the monomer liquid can flow out of the at least one ampoule into the cavity. If the inlet channel, in particular the inlet channel end opposite the mixing unit, is arranged closer to the pivot point than the outlet channel, in particular the outlet channel end opposite the mixing unit, the monomer liquid, with substantially perpendicular spatial orientation of the device and without further intervention of a user of the device, will flow substantially completely through the inlet channel into the mixing unit, in particular the distal part of the interior space of the mixing unit, while at the same time the outlet channel will remain substantially free of the mixing unit, thus permitting improved discharge of the gas displaced from the interior space. This arrangement of inlet channel and outlet channel relative to one another thus improves mass transfer between mixing unit and reservoir.

(65) The inlet channel and the outlet channel can open into the interior space, in particular the distal part of the interior space, at the same spatial height along a longitudinal axis of the device, in particular the mixing unit. In this embodiment, the inlet channel and the outlet channel open into the distal part of the interior space of the mixing unit in the broadest sense “next to one another”.

(66) An embodiment of the device is characterized in that the inlet channel opens into the interior space, in particular into the distal part of the interior space, proximal to the inlet channel. The outlet channel, in particular an outlet channel end facing the mixing unit, is thus closer to the discharge piston than the inlet channel, in particular an inlet channel end facing the mixing unit. Thus, the inlet channel, in particular the inlet channel end facing the mixing unit, opens closer to the conveying piston than the outlet channel, in particular the outlet channel end facing the mixing unit. This reduces the risk of the monomer liquid flowing out of the interior space through the outlet channel directly after flowing into the interior space through the inlet channel, particularly if, as is preferred due to the design of the device being as compact as possible, the inlet channel and the outlet channel open into the distal part of the interior space close to one another. Preferably, the outlet channel and the inlet channel open into the interior space no further than 1 cm from one another.

(67) In order to ensure that the monomer liquid is conveyed from the reservoir, preferably from the at least one ampoule opened in a fluid-conducting manner stored in the reservoir, substantially completely via the inlet channel, and not via the outlet channel, into the distal part of the interior space, an embodiment of the device is characterized in that the inlet channel is formed as a funnel

at the inlet channel end opposite the mixing unit and facing the reservoir. The design as a funnel facilitates the flow of the monomer liquid into the inlet channel. In particular, when storing the monomer liquid in an ampoule, in particular a glass ampoule, the shaping as a funnel can be advantageous, since the monomer liquid may flow out of the at least one ampoule opened in a fluid-conducting manner irregularly, for example intermittently. The design as a funnel improves the reception of the monomer liquid into the inlet channel, in particular in the case of irregular flow. For example, the funnel can assume a diameter in a range of 1 cm to 4 cm. The cross-sectional area of the funnel can be designed round, angular or elliptical, for example.
(68) The implementation as a funnel can take up different proportions of the total length of the inlet channel. For example, the inlet channel can be designed as a funnel over 10% to 95%, preferably 30% to 90%, more preferably 50% to 90%, of the total length of the inlet channel.
(69) The outlet channel and the inlet channel may have different diameters relative to one another.
(70) An embodiment of the device is characterized in that the outlet channel has a minimum outlet channel diameter that is at least half the minimum inlet channel diameter of the inlet channel. The minimum inlet channel diameter is thus preferably at most twice as large as the minimum outlet channel diameter.
(71) This ensures that gas from the interior space can be discharged into the reservoir through the outlet port sufficiently quickly so as not to slow the conveying of the monomer liquid from the reservoir into the interior space.
(72) In a preferred embodiment, the minimum outlet channel diameter and the minimum inlet channel diameter are substantially equal.
(73) The inlet channel and the outlet channel can be designed in one piece. Thus, they can be formed from a single, continuous component.
(74) An embodiment of the device is characterized in that the inlet channel and the outlet channel are formed at least in two parts, preferably in two or three parts, further preferably in two parts, such that the mixing unit, in particular the distal part of the interior space of the mixing unit, and the reservoir, preferably the cavity of the reservoir, are connected to one another in a fluid-conducting manner in a first channel position of the inlet channel and the outlet channel and are separated from one another in terms of fluid conduction in a second channel position of the inlet channel and the outlet channel. Preferably, the inlet channel and the outlet channel are reversible to the first channel position and the second channel position. A one-piece channel cannot reversibly assume two channel positions.
(75) In this embodiment, at least two components, preferably two or three components, more preferably two components, are involved in forming the inlet channel and the outlet channel, and the inlet channel and the outlet channel extend through these components. For example, the inlet channel is formed by connecting two disjoint tubes in a fluid-conducting manner.
(76) Preferably, the part of the channels that faces the mixing unit extends in one of the components and the part of the channels that faces the reservoir extends in another component.
(77) Preferably, the inlet channel and the outlet channel are formed by the same components.
(78) Due to the at least two-part design of inlet channel and outlet channel, both channels can establish a fluid-conducting connection between the reservoir and the mixing unit, preferably reversibly, in a first channel position and separate the reservoir and the mixing unit in a second channel position in a fluid-conducting manner. For example, the inlet channel is formed by connecting two previously disjoint tubes in a fluid-conducting manner, wherein the tubes connected to one another in a fluid-conducting manner represent the first channel position and the two disjoint tubes represent the second channel position.
(79) The at least two-part design of the inlet channel and the outlet channel thus allows a user of the device to control the mass transfer between the mixing unit and the reservoir.
(80) Preferably, the inlet channel and the outlet channel can be moved to the first or second channel position simultaneously.

(81) An embodiment of the device is characterized in that the device comprises a closure element which closes or makes closable in terms of fluid conduction, preferably reversibly closes or makes closable in terms of fluid conduction, at least a part of the at least two-part inlet channel that faces the mixing unit, i.e., adjacent to the mixing unit, and of the a part of the at least two-part outlet channel that faces the mixing unit, i.e., adjacent to the mixing unit. For example, the device comprises a plug reversibly insertable into the inlet channel end and outlet channel end facing the mixing unit to separate the mixing unit and the reservoir in terms of fluid conduction.
(82) The closure element facilitates substantially complete conveying of the monomer liquid from the distal part of the interior space through the conduit means into the proximal part of the interior space. Without the closure element, the monomer liquid, or portions thereof, could be discharged back out of the mixing unit as it is conveyed into the proximal part of the interior space through the inlet channel and/or the outlet channel.
(83) An embodiment of the device is characterized in that the mixing unit and the reservoir are reversibly connected or connectable to one another via a first form closure. Thus, the mixing unit and the reservoir are designed to be reversibly separable from one another, which simplifies the use of the device for a user. This allows easy separation of the reservoir, which is no longer required after the monomer liquid has been conveyed into the mixing unit, in particular into the distal part of the interior space of the mixing unit, thus simplifying the provision, and preferably the discharge, of the bone cement paste by means of the mixing unit, in particular by improving manageability.
(84) Preferably, the same components that form the at least two-part inlet channel and the at least two-part outlet channel are involved in the formation of the first form closure.
(85) In an embodiment, the inlet channel and the outlet channel are in the first channel position when the first form closure is formed and in the second channel position when the first form closure is released.
(86) The closure element can be shaped differently to separate the mixing unit and the reservoir in terms of fluid conduction.
(87) An embodiment of the device is characterized in that the closure element is a rotary valve through which the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit, i.e. the part of the two-part inlet channel and the two-part outlet channel adjacent to the mixing unit, extends and which, in a first rotary valve position, leaves the inlet channel and the outlet channel in the first channel position and, by rotating to a second rotary valve position, moves the inlet channel and the outlet channel to the second channel position. The part of the at least two-part inlet channel and two-part outlet channel facing away from the mixing unit and toward the reservoir does not extend through the rotary valve in each case. The rotary valve thus forms the component of the at least two-part inlet channel and outlet channel which forms the parts of the two channels that face the mixing unit. When the rotary valve is in the first rotary valve position, the inlet channel and the outlet channel are in the first channel position, thereby connecting the mixing unit, in particular the distal part of the interior space of the mixing unit, and the reservoir in a fluid-conducting manner. When the rotary valve is in the second rotary valve position, the inlet channel and the outlet channel are in the second channel position, thereby separating the mixing unit, in particular the distal part of the interior space of the mixing unit, and the reservoir in terms of fluid conduction. By rotating the rotary valve to the second rotary valve position, and thus moving the inlet channel and the outlet channel to the second channel position, the two parts of the inlet channel and the two parts of the outlet channel are spatially displaced relative to one another such that mass transfer between the respective parts of the inlet channel and the outlet channel is prevented. By rotating the rotary valve to the first rotary valve position, and thus moving the inlet channel and the outlet channel to the first channel position, the two parts of the inlet channel and the two parts of the outlet channel are arranged relative to one another in such a manner as to allow mass transfer between the corresponding parts of the inlet channel and the outlet channel.
(88) An embodiment of the device is characterized in that, after the reservoir has been separated

from the mixing unit by releasing the first form closure, the closure element can be brought into a closure position in order to close, in terms of fluid conduction, the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit. In this embodiment, the parts of the at least two-part inlet channel and the two-part outlet channel are separated in terms of fluid conduction by releasing the first form closure, wherein preferably in each case the part of the inlet channel and of the outlet channel that faces the reservoir, i.e., adjacent to the reservoir, together with the reservoir are removed from the mixing unit by releasing the first form closure, and the parts of the inlet channel and of the outlet channel adjacent to the mixing unit remain on the mixing unit, for example as feedthroughs in the cartridge wall in the region of the distal part of the interior space. In this embodiment, releasing the first form closure moves the inlet channel and the outlet channel from the first channel position to the second channel position. The closure element is used to close the parts of the inlet channel and the outlet channel remaining on the mixing unit, preferably from outside the mixing unit, after the reservoir has been removed. For this purpose, the closure element is brought into the closure position. For example, the closure element is a plug which can be brought into the closure position by pushing it into the part of the inlet channel and outlet channel remaining on the mixing unit, preferably from outside the mixing unit.
(89) An embodiment of the device is characterized in that the closure element is a screw. The screw preferably comprises an external thread which cooperates with an internal thread, which is preferably located outside the mixing unit, such that, after loosening the first form closure and removing the part of the inlet channel and the outlet channel facing the reservoir, screwing in the screw in the direction of the cartridge wall causes fluid-conducting closure of the part of the inlet channel and the outlet channel remaining on the mixing unit, preferably by means of a screw syringe.
(90) Preferably, the screw comprises wings to facilitate screwing to the locking position by a user of the device.
(91) The first form closure may be the only mechanical connection between the mixing unit and the reservoir.
(92) An embodiment of the device is characterized in that the mixing unit and the reservoir are reversibly connected or connectable to one another, in addition to the first form closure, via a second form closure.
(93) In this embodiment, the mixing unit and the reservoir are connected to each other by two form closures.
(94) Upon opening the at least one ampoule by tilting about the pivot point against the connection, a sufficiently large force must be exerted on the ampoule in order to overcome the structural integrity of the ampoule. This force is additionally increased given use of more than one ampoule, such as preferably two ampoules, if the latter are to be opened simultaneously, as is preferred, by tilting about the pivot point against the connection. This force acts on the contact points of the reservoir and mixing unit.
(95) If the reservoir and the mixing unit were connected only via the first form closure, the force for opening the at least one ampoule would act completely on this first form closure, such that, since preferably the components which form the at least two-part inlet channel and the at least two-part outlet channel are involved in the first form closure, kinking or even tearing of the inlet channel and/or the outlet channel could occur. This would make it difficult, if not impossible, to convey substantially all of the monomer liquid from the reservoir into the mixing unit, in particular the distal part of the interior space.
(96) The second form closure between the connecting element and the mixing unit ensures a force distribution of the force, required to open the at least one ampoule, onto the two form closures so that the risk of the device being damaged upon opening the at least one ampoule by tilting is reduced.
(97) Via the two form closures, the reservoir is connected stably to the mixing unit in such a way

that the at least one ampoule can be opened by tilting about the pivot point against the connection without damaging the device, and without requiring additional aids in addition to the device to open the at least one ampoule.
(98) The second form closure between the mixing unit and the reservoir can be realized in different ways.
(99) In an embodiment of the device, the second form closure is formed by means of a clasp. Preferably, the clasp is formed of two clasp indentations on a reservoir outer surface, such as an outer surface of the reservoir container, cavity or joint, and two clasp protrusions on a mixing element outer surface, preferably a cartridge outer surface, wherein the two clasp protrusions are reversibly insertable into the two clasp indentations to form the second form closure. This allows a quickly and easily creatable and releasable second form closure, which is stable and allows safe opening of the at least one ampoule by tilting around the pivot point.
(100) The pivot point as well as the first form closure and the second form closure may be arranged spatially in different ways with respect to one another.
(101) One embodiment of the device is characterized in that, in a side view, in particular a side view of the device, the pivot point, the first form closure, and the second form closure form the vertices of a triangle. In this embodiment, the pivot point and the two form closures lie in a common plane but are not arranged on a straight line running in this plane. The longitudinal axis of the cartridge preferably lies within this plane or runs at least parallel to this plane. The arrangement in the form of a triangle improves the force distribution of the force required for the opening of the at least one ampoule in terms of fluid conduction, in particular given a tilting movement, used for this purpose, of the ampoule about the pivot point within or parallel to the plane of the triangle. Furthermore, such an arrangement fixes and does not displace the pivot point in this plane, which facilitates a reproducible opening of the at least one ampoule.
(102) The pivot point as well as the first form closure and the second form closure may have different distances from one another.
(103) One embodiment of the device is characterized in that the second form closure has a shorter distance from the first form closure than the pivot point. In this embodiment, the distance between pivot point and first form closure is thus greater than the distance between second form closure and first form closure. In particular given an arrangement of pivot point and the two form closures in the form of a triangle, this allows both improved force distribution of the force, required during tilting to open the ampoule, onto the two form closures and simultaneously an optimally space-saving design of the device. The latter facilitates in particular the handling of the device by a user.
(104) The pivot point and the two form closures in this case preferably form the vertices of a triangle in a side view, wherein the pivot point and the second form closure have the smallest value of the three possible distances between the mentioned points. This leads to a further improvement of the force distribution onto the two form closures and allows a further space-saving design of the device.
(105) One embodiment of the device is characterized in that the first form closure, the second form closure, and the pivot point respectively lie on a straight line running parallel to a longitudinal axis of the cartridge, wherein the straight lines have a different straight-line distance from the longitudinal axis of the cartridge. This arrangement improves the force distribution of the force, required during tilting to open the ampoule, onto the two form closures and also simultaneously allows an optimally space-saving design of the device. The latter facilitates in particular the handling of the device by a user.
(106) The pivot point and the two form closures in this case preferably form the vertices of a triangle, wherein the triangle lies in a plane in which the longitudinal axis of the cartridge also lies, or to which the longitudinal axis of the cartridge runs at least in parallel.
(107) A further subject matter of the invention relates to a method for providing a bone cement paste from two starting components by means of a device, in particular by means of a device

according to any of the preceding embodiments, comprising the following steps: a. the monomer liquid flowing from the reservoir through the inlet channel into the distal part of the interior space while simultaneously a gas from the interior space is discharged through the outlet channel into the reservoir, b. conveying the monomer liquid from the distal part of the interior space through the conduit means into the proximal part of the interior space by means of advancing the conveying piston in the direction of the discharge piston.

(108) Preferably, the flow of the monomer liquid in step a. occurs in accordance to gravity. For this purpose, the device is held spatially by a user in such a manner that the discharge piston is arranged spatially above the conveying piston, preferably perpendicular above the conveying piston.

(109) As the monomer liquid flows through the inlet channel into the distal part of the interior space, a temporary monomer liquid level may be formed within the inlet channel, which results from an inlet volume of monomer liquid into the inlet channel and an outlet volume of monomer liquid from the inlet channel into the distal part of the interior space. By the aforementioned embodiments of the device, the monomer liquid level is preferably always formed distal to the, reservoir-facing, exit channel end of the exit channel such that when the device is oriented perpendicular with the proximal cartridge end facing upward, substantially no monomer liquid enters the exit channel on the reservoir side.

(110) Driving the conveying piston forward in the direction of the discharge piston in step b. reduces the spatial distance between the two pistons such that, depending on the volume of the monomer liquid present in the distal part of the interior space, at a determined spatial proximity the monomer liquid is conveyed from the distal part of the interior space through the conduit means into the proximal part of the interior space.

(111) With the beginning of conveying the monomer liquid into the proximal part of the interior space, there is a contact of monomer liquid in the bone cement powder stored in the proximal part of the interior space, which is accompanied by formation of the bone cement paste.

(112) An embodiment of the method is characterized in that, before the monomer liquid is conveyed in step b., the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit is closed in terms of fluid conduction by the closure element.

(113) This facilitates substantially complete conveying of the monomer liquid from the distal part of the interior space through the conduit means into the proximal part of the interior space without substantial portions of the monomer liquid being discharged from the mixing unit from the distal part of the interior space through the inlet channel and/or the outlet channel by advancing the conveying piston. Furthermore, a user does not need to spatially orient the device precisely to substantially eliminate this discharge.

(114) The conveying piston can be advanced in different manners in the direction of the discharge piston. For example, a user of the device can advance the conveying piston manually, in particular by applying force to a rod or axle. In a further embodiment, the cartridge and the conveying piston together form a thread via which the conveying piston can be screwed into the cartridge in the direction of the discharge piston. Preferably, the cartridge comprises an internal thread and the conveying piston comprises an external thread, which interact positively and/or non-positively to enable the conveying piston to be driven forward.

(115) In a further embodiment of the method, the conveying piston is advanced using a mechanical aid.

(116) An embodiment of the method is characterized in that for advancing the conveying piston, the device is inserted into a discharge device, in particular a discharge gun for bone cement paste. Discharge guns for bone cement paste are known to the person skilled in the art.

(117) As the monomer liquid is conveyed from the rear part to the front part of the interior space, the formation of the bone cement paste from the two starting components begins. Preferably, this is carried out by mixing the two starting components as uniformly as possible to obtain a bone cement paste that is as homogeneous as possible. The mixing of the two starting components can

be done in different ways. In an embodiment of the method, mixing is carried out with the active participation of the user of the device, for example by shaking the device or by actuating a mixing element in the front part of the interior space, in particular a stirring device.
(118) An embodiment of the method is characterized in that the monomer liquid is distributed in the bone cement powder with the aid of a hydrophilic additive. An advantage is that this takes place without the active participation of the user of the device, which avoids possible errors by the user during mixing. A possible error is that the user does not mix over the entire length of the front part of the interior space, such that parts of the bone cement powder are not wetted with monomer liquid. A further advantage is that the device can thus be designed more simply and with fewer moving parts, which reduces both the risk of malfunctions and the manufacturing costs of the device.
(119) The device is characterized in that it provides a bone cement paste from two starting components. Bone cement paste is understood to mean a substance that is suitable in the field of medical technology for creating a stable connection between artificial joints, such as hip and knee joints, and bone material. By curing, a bone cement paste becomes a bone cement. These bone cements are preferably polymethyl methacrylate bone cements (PMMA bone cements). PMMA bone cements have been used for a long time in medical applications and are based upon the work of Sir Charnley (cf. Charnley, J., Anchorage of the femoral head prosthesis of the shaft of the femur. *J. Bone Joint Surg.* 1960; 42, 28-30.). PMMA bone cements can thereby be produced from a bone cement powder as a first starting component and a monomer liquid as a second starting component. With a suitable composition, the two starting components can be storage-stable, separately from one another. When the two starting components are brought into contact with one another, a plastically-deformable bone cement paste is produced by the swelling of the polymer components of the bone cement powder. In this case, polymerization of the monomer by radicals is initiated. As the polymerization of the monomer progresses, the viscosity of the bone cement paste increases until it cures completely.
(120) Bone cement powder is understood to mean a powder that comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer. Examples of copolymers are styrene and/or methyl acrylate. In one embodiment, the bone cement powder can additionally comprise a hydrophilic additive which supports the distribution of the monomer liquid within the bone cement powder. In a further embodiment, the bone cement powder can additionally comprise an initiator which initiates the polymerization. In a further embodiment, the bone cement powder can additionally comprise a radiopaque material. In yet another embodiment, the bone cement powder can additionally comprise pharmaceutically-active substances, such as antibiotics.
(121) The bone cement powder preferably comprises, as a hydrophilic additive, at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, and a radiopaque material, or consists of these components. More preferably, the bone cement powder comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, a radiopaque material, and a hydrophilic additive, or consists of these components. Most preferably, the bone cement powder comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, a radiopaque material, a hydrophilic additive, and an antibiotic, or consists of these components.
(122) According to the invention, the particle size of the particulate polymethyl methacrylate and/or of the particulate polymethyl methacrylate copolymer of the bone cement powder can correspond to the sieve fraction of less than 150 μm, preferably less than 100 μm.
(123) According to the invention, the hydrophilic additive can be designed in particulate and/or fibrous form. In a further embodiment, the hydrophilic additive can be slightly soluble, and preferably insoluble, in methyl methacrylate. In a further embodiment, the hydrophilic additive may have an absorption capacity of at least 0.6 g of methyl methacrylate per gram of hydrophilic

additive. In a further embodiment, the hydrophilic additive can comprise a chemical substance comprising at least one OH group. In this case, the hydrophilic additive can preferably have covalently-bonded OH groups at its surface. Examples of such preferred hydrophilic additives can be additives selected from the group comprising cellulose, oxycellulose, starch, titanium dioxide, and silicon dioxide, wherein pyrogenic silicon dioxide is particularly preferred. In one embodiment, the particle size of the hydrophilic additive can correspond to the sieve fraction of less than 100 μm, preferably less than 50 μm, and most preferably less than 10 μm. The hydrophilic additive can be contained in an amount of 0.1 to 2.5% by weight, based on the total weight of the bone cement powder.
(124) According to the invention, the initiator can contain dibenzoyl peroxide or consist of dibenzoyl peroxide.
(125) According to the invention, a radiopaque material is understood to mean a substance that makes it possible to make the bone cement visible on diagnostic X-ray images. Examples of radiopaque materials can include barium sulfate, zirconium dioxide, and calcium carbonate.
(126) According to the invention, the pharmaceutically-active substance can comprise one or more antibiotics and, optionally, added cofactors for the one or more antibiotics. Preferably, the pharmaceutically-active substance consists of one or more antibiotics and, optionally, added cofactors for the one or more antibiotics. Examples of antibiotics include, inter alia, gentamicin, clindamycin, and vancomycin.
(127) According to the invention, the monomer liquid can comprise the monomer methyl methacrylate or consist of methyl methacrylate. In one embodiment, the monomer liquid comprises, in addition to the monomer, an activator dissolved therein, such as N,N-dimethyl-p-toluidine, or consists of methyl methacrylate and N,N-dimethyl-p-toluidine.
(128) The features disclosed for the device are also disclosed for the method, and vice versa.

Description

FIGURES
(1) In the following, the invention is illustrated further, by way of example, by figures. The invention is not limited to the figures.
(2) In the figures:
(3) FIG. **1** shows a schematic longitudinal section of an exemplary device for providing a bone cement paste comprising a mixing unit and a reservoir with an ampoule filled with a monomer liquid,
(4) FIG. **2** shows the device of FIG. **1**, wherein a first form closure, a second form closure, and a pivot point are indicated,
(5) FIG. **3** shows a perspective side view of the mixing unit of the device from FIGS. **1** and **2**,
(6) FIG. **4** shows the device of FIGS. **1** to **3** during a fluid-conducting opening of the ampoule and during conveying of the monomer liquid into the mixing unit,
(7) FIG. **5** shows the device of FIGS. **1** to **4**, with the reservoir being separate from the mixing unit,
(8) FIG. **6** shows the device from FIGS. **1** to **5** with a closure element closed in terms of fluid conduction in the form of a screw,
(9) FIG. **7** shows the device of FIGS. **1** to **6** during conveying the monomer liquid into the bone cement powder,
(10) FIG. **8** shows the device from FIGS. **1** to **7** with bone cement paste provided,
(11) FIG. **9** shows the device from FIGS. **1** to **8** during discharge of the bone cement paste,
(12) FIG. **10** shows a schematic longitudinal section of a further exemplary device for providing a bone cement paste comprising a mixing unit and a reservoir with an ampoule filled with a monomer liquid,

(13) FIG. **11** shows the device of FIG. **10** during a fluid-conducting opening of the ampoule and during conveying of the monomer liquid into the mixing unit,
(14) FIG. **12** shows the device of FIGS. **10** and **11** with a closure element closed in terms of fluid conduction in the form of a rotary valve, and
(15) FIG. **13** shows a flow diagram of a method for providing a bone cement paste.
DESCRIPTION OF THE FIGURES
(16) FIG. **1** shows a schematic longitudinal section of an exemplary embodiment of a device **100** for providing a bone cement paste from two starting components in an initial state. The device **100** comprises a mixing unit **200** and a reservoir **300**, which are reversibly separably connected to one another via a first form closure and a second form closure (see FIGS. **2** and **3**).
(17) The mixing unit **200** is of tubular construction and comprises a hollow cylindrical cartridge **210** having an interior space **215**, which is divided into a proximal part **216** and a distal part **217** of the interior space **215** by a discharge piston **220** that is reversibly axially displaceable in the interior space **215**. The proximal part **216** of the interior space **215** stores a bone cement powder **500** as a first starting component of the bone cement paste, and the reservoir **300** stores an ampoule **330** containing a monomer liquid **510** as a second starting component of the bone cement paste. The bone cement powder **500** contains particulate polymethyl methacrylate as the main component, as well as a hydrophilic additive that allows the monomer liquid **510** to be dispersed in the bone cement powder **500** without mixing.
(18) The discharge piston **220** is designed to be impermeable to solids such that no bone cement powder **500** can pass from the proximal part **216** into the distal part **216** of the interior space **215**. The discharge piston **220** comprises a conduit means **230** (marked only by way of example) in the form of a plurality of passages through which a fluid-conducting connection is formed between the proximal part **216** and the distal part **217** of the interior space **215**. The conduit means **230** is impermeably closed to solids or bone cement paste by a filter means **235** in the form of a pore disc, wherein the pore disc allows conveying of the monomer liquid **510** from the distal part **217** into the proximal part **216** of the interior space **215** without any problems. In the embodiment of the device **100** shown, the filter means **235** is disposed on the proximal end of the conduit means **230** facing the proximal part **216** of the interior space **215**. In further embodiments, not shown, the filter means **235** is disposed on the distal end of the conduit means **230** facing the distal part **217** of the interior space **215**, or on both ends of the conduit means **230**. An advantage of a filter means **235** arranged as shown is that the bone cement paste forming in the proximal part **216** of the interior space **215** cannot clog the conduit means **230**.
(19) A conveying piston **240** axially movable within the interior space **215** is arranged distal to the discharge piston **220** within the distal part **217** of the interior space **215**. The conveying piston **240** closes a distal cartridge end **212** of the cartridge **210** in terms of fluid conduction.
(20) The mixing unit **200** further comprises, at a proximal cartridge end **211** opposite the distal cartridge end **212**, a discharge opening **250** that delimits the region of the proximal part **216** of the interior space **215** of the cartridge **210** facing away from the discharge piston **220**. In the initial state of the device **100**, the discharge opening **250** is closed by a closure cap **260** with a plug **265** such that no bone cement powder **500** can escape from the cartridge **210**. The plug **265** is designed to be gas permeable to allow a gas **530** present in the interior space **215** to transfer through the bone cement powder **500** and out of the device **100** before formation of the bone cement paste begins.
(21) The reservoir **300** comprises a tubular reservoir container **310** in which two ampoules **330**, in particular two glass ampoules, are stored side by side (only one of the ampoules **330** is visible in the view shown). The ampoules **330** each comprise an ampoule body **331**, an ampoule head **332** facing the mixing unit **200**, and an ampoule neck **333** located between the ampoule body **331** and the ampoule head **332**, which acts as a predetermined breaking point for the ampoules **330**. The monomer liquid **510** is stored in the ampoules **330**. The ampoule head **332** of the ampoule **330** is arranged in portions in a connection **350** which connects a cavity **340** of the reservoir **300** to the

ampoule **330**. The connection **350** has a connection diameter **355** that is about 5% larger than a diameter of the ampoule heads **332**, such that the connection **350** fixes the ampoule heads **332** against tilting within the drawing plane. To allow tilting of the ampoules **330**, in particular the ampoule heads **332**, against the connection **350**, wherein in the embodiment shown tilting in the drawing plane is possible, the reservoir container **310** comprises a deformable region **320** in the region of a transition from the connection **350** to the ampoule body **331**.

(22) Within the cavity **340**, a filter element **345** is arranged in the reservoir **300** so that after the ampoules **330** have been opened in terms of fluid conduction, fragments thereof cannot pass via the cavity **340** into the mixing unit **200** but rather are retained on the filter element **345**.

(23) The reservoir **300**, in particular the cavity **340**, is connected to the mixing unit **200**, in particular the distal part **217** of the interior space **215**, via an inlet channel **400** and an outlet channel **410** in a fluid-conducting manner. In the view shown, the inlet channel **400** and the outlet channel **410** are in a first channel position such that the reservoir **300** and the mixing unit **200** are connected to one another via the two channels **400**, **410** in a fluid-conducting manner.

(24) The inlet channel **400** is used to introduce the monomer liquid **510** from the reservoir **300** into the mixing unit. The outlet channel serves to discharge the gas **530** from the mixing unit **200** into the reservoir **300**, which is displaced by the monomer liquid **510** flowing into the distal part **217** of the interior space **215**. To ensure good mass transfer between the reservoir **300** and the mixing unit **200**, the outlet channel **410** in the embodiment shown has a minimum outlet channel diameter **411**, which corresponds to a minimum inlet channel diameter **401** of the inlet channel **400**. The inlet channel **400** is shaped as a funnel at an end facing the ampoules **330** to better receive the monomer liquid **510** as it flows out of the ampoules **330** to be opened.

(25) Both the inlet channel **400** and the outlet channel **410** are formed in two parts, wherein a part of the inlet channel **400** and of the outlet channel **410** facing the mixing unit **200** is respectively formed as a feedthrough in the cartridge **210**, and a part of the inlet channel **400** and of the outlet channel **410** facing the ampoules **330** respectively extends within the reservoir **300** that can be reversibly separated from the mixing unit **200**. Thus, the inlet channel **400** and the outlet channel **410** are shaped from two components of the device **100**.

(26) The device **100** further comprises a closure element **450** in the form of a screw which, after separation of the mixing unit **200** and the reservoir **300**, can close, in terms of fluid conduction, the parts of the inlet channel **400** and the outlet channel **410** that face the mixing unit **200**, i.e., the feedthroughs in the cartridge **210** (cf. FIG. **6**).

(27) FIG. **2** shows the device **100** of FIG. **1**, wherein the first form closure **430**, the second form closure **440**, and a pivot point **420** about which, due to the deformable region **320**, the ampoules **330** can be pressed by tilting against the connection **350** (cf. FIG. **1**), are indicated by filled circles.

(28) The first form closure **430** is formed by the components shaping the inlet channel **400** and the outlet channel **410** (cf. FIG. **1**) in the region of the inlet channel **400** and the outlet channel **410**. The second form closure **430** is designed by a clasp (not shown in FIGS. **1** and **2**; cf. FIG. **3**).

(29) In the shown side view of the device **100**, the first form closure **430**, the second form closure **440**, and the pivot point **420** form a triangle (indicated by connecting lines between the filled circles). When the ampoules **330** are tilted about the pivot point **420** such that the ampoules **330**, in particular the ampoule heads **332** (cf. FIG. **1**), are pressed against the connection **350** (cf. FIG. **1**), the force required in this case to open the ampoules **330** in a fluid-conducting manner is distributed between the first form closure **430** and the second form closure **440**. An advantageous force distribution is achieved via the arrangement of the first form closure **430**, the second form closure **440**, and the pivot point **420** in the form of a triangle. In particular, this can reduce the risk of kinking or breaking of the inlet channel **400** and outlet channel **410**.

(30) The first form closure **430**, the second form closure **440**, and the pivot point **420** are arranged relative to one another such that the second form closure **440** has a shorter distance from the first form closure **430** than the pivot point **420**. The pivot point **420** and the first form closure **430** are

thus spaced farther apart from one another than the first form closure **430** and the second form closure **440**. This ensures improved force distribution of the force required during tilting about the pivot point **420** to open the ampoules **330** onto the two form closures **430**, **440** and also, at the same time, ensures an optimally space-saving design of the device **100**. The latter in particular facilitates the handling of the device **100** by a user.

(31) The first form closure **430** lies on a straight line **431** extending parallel to a longitudinal axis **213** of the cartridge **210**, the second form closure **440** lies on another straight line **441** extending parallel to the longitudinal axis **213** of the cartridge **210**, and the pivot point **420** lies on another straight line **421** extending parallel to the longitudinal axis **213** of the cartridge **210**, wherein the straight lines **431**, **441**, **421** all have a different straight-line distance from the longitudinal axis **213** of the cartridge **210**. In the embodiment shown, the straight-line distance between the straight line **421** through the pivot point **420** and the longitudinal axis **213** is greatest, followed by the straight-line distance between the straight line **431** through the first form closure **430** and the longitudinal axis **213**. The different straight-line distances improve the force distribution of the force required during tilting about the pivot point **420** to open the ampoules **330** onto the two form closures **430**, **440**, and at the same time allow an optimally space-saving design of the device **100**. The latter in particular facilitates the handling of the device **100** by a user.

(32) FIG. **3** shows a perspective side view of the mixing unit **200** of the device **100** of FIGS. **1** and **2**. The mixing unit **200** comprises two clasp protrusions **261** on an outer surface of the cartridge **210**, which are reversibly insertable into clasp indentations on an outer surface of the reservoir **300** (not shown) to form a clasp (not shown) that forms the second form closure (see FIG. **2**).

(33) FIG. **4** shows the device **100** of FIGS. **1** to **3** with ampoules **330** tilted about pivot point **420** (see FIG. **2**). For this purpose, the reservoir container **310** is bent at the deformable region **320** such that the ampoule heads **332** were pressed against the connection **350** and the ampoules **330** were opened in the region of the ampoule neck **332** (cf. FIG. **1**) in a fluid-conducting manner. Most of the monomer liquid **510** stored in the ampoules **330** opened in a fluid-conducting manner has already flowed out of the ampoules **330** via the cavity **340** and the inlet channel **400** into the distal part **217** of the interior space **215**. The ampoule head **332** of one of the ampoules **330** has completely transitioned from the connection **350** into the cavity **340**. In this case, the ampoule head **332** has been captured by the filter element **345** such that it, or respectively fragments thereof, cannot pass to or through the inlet channel **400**. The cavity **340** is dimensioned such that the ampoule head **332** can be mounted so as to be completely rotatable therein, so that monomer liquid **510** that is possibly still present in the ampoule head **332** after the opening of the ampoule **330** can flow out into the cavity **340**. This has already happened in FIG. **4**. At the same time, a portion of the gas **530** has been discharged from the interior space **215** into the reservoir **300** via the outlet channel **410**. The volume of the discharged portion of the gas **530** is substantially equal to the volume of the monomer liquid **510** already introduced into the distal part **217** of the interior space **215**.

(34) In order to convey substantially all of the monomer liquid **510** into the mixing unit **200** through the inlet channel **400**, rather than through the outlet channel **410**, the inlet channel **400** has a smaller distance to the pivot point **420** than the outlet channel **410**. Furthermore, the outlet channel **410** opens into the interior space **215** proximal to the inlet channel **400** such that the inflowing monomer liquid **510** does not impede discharge of the gas **530** from the interior space **215** through the outlet channel **410**.

(35) FIG. **5** shows the device **100** of FIGS. **1** to **4**, wherein the reservoir **300** has been separated from the mixing unit **200** by releasing the first form closure **430** and the second form closure **440** after substantially complete flow of the monomer liquid **510** into the distal part **217** of the interior space **215**. By separating the reservoir **300**, the parts of the inlet channel **400** and the outlet channel **410** facing the reservoir **300** were separated in terms of fluid conduction from the parts of the inlet channel **400** and outlet channel **410** facing the mixing unit **200**. The inlet channel **400** and the outlet

channel **410** are thus in a second channel position separated in terms of fluid conduction.
(36) The distal part **217** of the interior space **215** is connected in a fluid-conducting manner to the surroundings of the device **100** via the part of the inlet channel **400** and outlet channel **410** that faces the mixing unit **200**, i.e., the feedthroughs in the cartridge **210**. Thus, advancing the conveying piston **240** in the direction of the discharge piston **220** would, in the embodiment of the device **100** shown, discharge the monomer liquid **510** at least partially from the mixing unit **200** instead of, as desired, conveying it substantially completely through the conduit means **230** into the proximal part **216** of the interior space **215** to therein form the bone cement paste together with the bone cement powder **500**.
(37) FIG. **6** shows the device **100** of FIGS. **1** to **6**, wherein the closure element **450** is screwed in until it contacts the cartridge **210**, and thus has been brought into a closure position in which the part of the inlet channel **400** and of the outlet channel **410** that faces the mixing unit **200** is closed off from the surroundings of the device **100** in terms of fluid conduction. In the closure position, the closure element **450** allows substantially complete conveying of the monomer liquid **510** into the proximal part **216** of the interior space by advancing the conveying piston **240** in the direction of the discharge piston **220**.
(38) To facilitate conveying of the monomer liquid **510** into the proximal part **216** of the interior space **215** for the user, the distal cartridge end **212** is connected to a discharge aid **550** in the form of a discharge gun that can push distally against the conveying piston **240** to displace it proximally.
(39) FIG. **7** shows the device **100** of FIGS. **1** to **6** during conveying of the monomer liquid **510** from the distal part **217** of the interior space **215** through the conduit means **230** into the proximal part **216** of the interior space **215**. In this regard, due to the sealing means **450** being in the sealing position, no monomer liquid **510** is released from the distal part **217** of the interior space **215** to the surroundings of the device **100**.
(40) As the monomer liquid **510** begins to come into contact with the bone cement powder **500**, the bone cement paste is formed in the proximal part **216** of the interior space **215**. In this context, the hydrophilic additive in the bone cement powder **500** ensures improved distribution of the monomer liquid **510** in the bone cement powder **500**, such that provision of a homogeneous bone cement paste is possible without mechanical aids, such as a mixing rod, and thus without mixing equipment.
(41) FIG. **8** shows the device **100** of FIGS. **1** to **7** with bone cement paste **520** provided in the proximal part **216** of the interior space **215**. For this purpose, the conveying piston **240** was displaced proximally by the discharge aid **550** until it abuts distally against the discharge piston **220**. This conveyed substantially all of the monomer liquid **510** into the proximal part **216** of the interior space **215**.
(42) The formation of the bone cement paste **520** has resulted in swelling of the bone cement powder **500**, which is accompanied by an increase in volume. As a result, the plug **265** has been deployed from the closure cap **260** in portions, signaling to the user of the device **100** that the bone cement paste **520** has been provided.
(43) FIG. **9** shows the device **100** of FIGS. **1** to **8** discharging the bone cement paste **520** from the discharge opening **250**. For this purpose, the conveying piston **240** together with the discharge piston **220** was advanced in the direction of the discharge opening **550** by a continued advancement of the discharge aid **550**. In further embodiments of the device **100** not shown, the discharge opening **250** may be provided with a discharge snorkel to assist in targeted discharge of the bone cement **520**.
(44) FIG. **10** shows another exemplary embodiment of a device **100′** for providing a bone cement paste from two starting components in an initial state. The embodiment of the device **100′** largely corresponds to the embodiment described above and shown in FIGS. **1** to **9**, and therefore reference is made to the above description in order to avoid repetitions. Modifications of any of the embodiments shown in FIGS. **1** to **9** have the same reference sign with an apostrophe.

(45) The device **100′** differs from the device **100** of FIGS. **1** to **9** in that it includes a closure element **450′** in the form of a rotary valve. The inlet channel **400′** and the outlet channel **410′** are designed in two parts, just as in the device **100** of FIGS. **1** to **9**, wherein each part of the two channels **400′**, **410′** that faces the mixing unit **200′** respectively extends through the closure element **450′**. In FIG. **10**, the closure element **450′** is in a first rotary valve position such that the inlet channel **400′** and the outlet channel **410′** are in the first channel position connecting the reservoir **300′** and the mixing unit **200′** to one another in a fluid-conducting manner.
(46) FIG. **11** shows the device **100′** of FIG. **10** with ampoules **330′** opened in a fluid-conducting manner.
(47) For this purpose, the reservoir container **310′** is bent at the deformable region **320′** such that the ampoule heads **332′** were pressed against the connection **350′** and the ampoules **330′** were opened in the region of the ampoule neck **332′** (cf. FIG. **10**) in a fluid-conducting manner. Most of the monomer liquid **510′** stored in the ampoules **330′** opened in a fluid-conducting manner has already flowed out of the ampoules **330′** via the cavity **340′** and the inlet channel **400′** into the distal part **217′** of the interior space **215′**. The ampoule head **332′** of one of the ampoules **330′** has completely transitioned from the connection **350′** into the cavity **340′**. In this case, the ampoule head **332′** has been captured by the filter element **345′** such that it, or respectively fragments thereof, cannot pass to or through the inlet channel **400′**. The cavity **340′** is dimensioned such that the ampoule head **332′** can be mounted so as to be completely rotatable therein, so that any monomer liquid **510′** that is possibly still present in the ampoule head **332′** after the opening of the ampoule **330′** can flow out into the cavity **340′**. This has already happened in FIG. **11**. At the same time, a portion of the gas **530′** has been discharged from the interior space **215′** into the reservoir **300′** via the outlet channel **410′**. The volume of the discharged portion of the gas **530′** is substantially equal to the volume of the monomer liquid **510′** already introduced into the distal part **217′** of the interior space **215′**.
(48) In order to convey substantially all of the monomer liquid **510′** into the mixing unit **200′** through the inlet channel **400′**, rather than through the outlet channel **410′**, the inlet channel **400′** has a smaller distance to the pivot point **420′** than the outlet channel **410′**. Furthermore, in the first rotary valve position, the outlet channel **410′** opens into the interior space **215′** proximally to the inlet channel **400′** such that the inflowing monomer liquid **510′** does not impede discharge of the gas **530′** from the interior space **215′** through the outlet channel **410′**.
(49) FIG. **12** shows the device **100′** of FIGS. **10** and **11** with monomer liquid **510′** conveyed substantially completely into the distal part **217′** of the interior space **215′**. To facilitate substantially complete conveying of the monomer liquid **510′** from the distal part **217′** of the interior space **215′** of the embodiment of the device **100′** shown, the closure element **450′** in the form of a rotary valve has been brought to a second rotary valve position by a rotation of about an axis of the closure element **450′**. The inlet channel **400′** and the outlet channel **410′** are in a second channel position in the second rotary valve position, in which the mixing unit **200′** and the reservoir **300′** are separated from one another in terms of fluid conduction. Conveying the monomer liquid **510′** into the proximal part **216′** of the interior space **215′** and providing and discharging the bone cement paste can be performed in a manner analogous to the device **100** as shown in FIGS. **7** to **9**.
(50) FIG. **13** illustrates a method **600** for providing a bone cement paste **520** from two starting components by means of the devices **100**, **100′** in accordance with FIGS. **1** to **9** and **10** to **12** comprising steps **610** and **620** and optionally step **615**.
(51) In a step **610**, the monomer liquid **510**, **510′** flows from the reservoir **300**, **300′** through the inlet channel **400**, **400′** into the distal part **217**, **217′** of the interior space **215**, **215′** of the mixing unit **200**, **200′**. At the same time, the monomer liquid **510**, **510′** flowing into the distal part **217**, **217′** of the interior space **215**, **215′** displaces, according to its volume, the gas **530**, **530′** present in the interior space **215**, **215′**, which is discharged from the interior space **215**, **215′** into the reservoir at the same time as the monomer liquid **510**, **510′** flows in. Thus, the inlet channel **400**, **400′**

synergistically interacts with the outlet channel **410**, **410′** to improve mass transfer between the reservoir **300**, **300′** and the mixing unit **200**, **200′**.

(52) In a step **620**, the monomer liquid **510**, **510′** is conveyed from the distal part **217**, **217′** of the interior space **215**, **215′** through the conduit means **230**, **230′** into the proximal part **216**, **216′** of the interior space **215**, **215′**. For this purpose, the conveying piston **240**, **240′** is advanced in the direction of the discharge piston **220**, **220′** within the interior space **215**, **215′**. In order to convey the monomer liquid **510**, **510′** substantially completely into the proximal part **216**, **216′** of the interior space **215**, **215′**, the conveying piston **240**, **240′** is preferably advanced in the direction of the discharge piston **220**, **220′** until it is in distal contact with the discharge piston **220**, **220′**.

(53) In a preferred embodiment of the method **600**, in a step **615**, which takes place between step **610** and step **620**, the part of the two-part inlet channel **400**, **400′** and the two-part outlet channel **410**, **410′** that faces the mixing unit **200**, **200′** is closed in terms of fluid conduction by the closure element **450**, **450′**. This facilitates a substantially complete conveying **620** of the monomer liquid **510**, **510′** into the proximal part **216**, **216′** of the interior space **215**, **215′** because the monomer liquid **510**, **510′** is not inadvertently discharged, by advancing the conveying piston **240**, **240′**, from the corresponding parts of the channels **400**, **400′**, **410**, **410′** of the device **100**, **100′**.

(54) Preferably, prior to step **620**, the reservoir **300**, **300′** is separated from the mixing unit **200**, **200′** by releasing the first form closure **430** and the second form closure **440**, which improves the handling of the device for a user.

(55) When conveying **620** the monomer liquid **510**, **510′** into the proximal part **216**, **216′** of the interior space **215**, **215′**, formation of the bone cement paste **520** from the bone cement powder **500**, **500′** and the monomer liquid **510**, **510′** occurs. Preferably, the formation of the bone cement paste **520** occurs without mechanical action by the user of the device **100**, **100′**. The method **600** is thus preferably carried out without mechanical action, for example without actuation of a mixing device, such as a mixing rod.

(56) After the bone cement paste **520** has been provided, it is preferably discharged from the device **100**, **100′**, in particular from the proximal part **216**, **216′** of the interior space **215**, **215′**, by a continued advancement of the conveying piston **240**, **240′** in the direction of the discharge opening **250**, **250′**. In the process, the conveying piston **240**, **240′** acts from a distal direction on the discharge piston **220**, **220′**, which is thereby moved in the direction of the discharge opening **250**, **250′**. Conveying **620** of the monomer liquid **510**, **510′** and discharging of the bone cement paste **520** is thus preferably carried out by means of unidirectional advancement of the conveying piston **240**, **240′**.

REFERENCE SIGNS

(57) **100**, **100′** device **200**, **200′** mixing unit **210**, **210′** hollow cylindrical cartridge **211**, **211′** proximal cartridge end **212**, **212′** distal cartridge end **213** longitudinal axis of the cartridge **215**, **215′** interior space of the cartridge **216**, **216′** proximal part of the interior space **217**, **217′** distal part of the interior space **220**, **220′** discharge piston **230**, **230′** conduit means **235**, **235′** filter means **240**, **240′** conveying piston **250**, **250′** discharge opening **260**, **260′** closure cap **265**, **265′** plug **300**, **300′** reservoir **310**, **310′** reservoir container **320**, **320′** deformable region **330**, **330′** ampoule **331**, **331′** ampoule body **332**, **332′** ampoule head **333**, **333′** ampoule neck **340**, **340′** cavity **345**, **345′** filter element **350**, **350′** connection **355**, **355′** connection diameter **361** clasp protrusions **400**, **400′** inlet channel **401**, **401′** inlet channel diameter **410**, **410′** outlet channel **411**, **411′** outlet channel diameter **420** pivot point **421** straight line through the pivot point **430** first form closure **431** straight line through the first form closure **440** second form closure **441** straight line through the second form closure **450**, **450′** closure element **500**, **500′** bone cement powder **510**, **510′** monomer liquid **520** bone cement paste **530**, **530′** gas **550**, **550′** discharge aid **600** method **610** flowing **615** closing **620** conveying

Claims

1. A device for providing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow cylindrical cartridge with an interior space, wherein a discharge piston axially movable in the interior space is arranged in the interior space, which discharge piston divides the interior space into a proximal part of the interior space and a distal part of the interior space, wherein the proximal part and the distal part of the interior space are connected to one another in a fluid-conducting manner via a conduit means, wherein a bone cement powder is stored in the proximal part of the interior space as a first starting component and wherein a conveying piston axially movable in the interior space is arranged in the distal part of the interior space, and a reservoir for a monomer liquid serving as a second starting component, which is connected or connectable in a fluid-conducting manner via an inlet channel to the distal part of the interior space for introducing the monomer liquid from the reservoir into the mixing unit, wherein the reservoir and the mixing unit are connected or connectable in a fluid-conducting manner via an outlet channel, via which a gas is discharged from the interior space into the reservoir.
2. The device according to claim 1, wherein the reservoir comprises a reservoir container in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head is arranged, and the monomer liquid is stored in the at least one ampoule, and comprises a cavity in the region of the ampoule head, wherein the cavity is connected to the inlet channel in a fluid-conducting manner and comprises a connection to the at least one ampoule, wherein the ampoule head is at least in regions disposed in the connection and the reservoir container comprises, at least in portions, a deformable region such that tilting of the at least one ampoule about a pivot point against the connection is enabled.
3. The device according to claim 2, wherein the inlet channel has a smaller distance to the pivot point than the outlet channel.
4. The device according to claim 1, wherein the outlet channel opens into the interior space proximally to the inlet channel.
5. The device according to claim 1, wherein the inlet channel is designed as a funnel at an inlet channel end opposite the mixing unit.
6. The device according to claim 1, wherein the outlet channel has a minimum outlet channel diameter corresponding to at least half a minimum inlet channel diameter of the inlet channel.
7. The device according to claim 1, wherein the inlet channel and the outlet channel are formed at least in two parts, such that the mixing unit and the reservoir are connected to one another in a fluid-conducting manner in a first channel position of the inlet channel and the outlet channel and are separated from one another in terms of fluid conduction in a second channel position of the inlet channel and the outlet channel.
8. The device according to claim 7, wherein the device comprises a closure element that closes or makes closable, in terms of fluid conduction, at least a part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit.
9. The device according to claim 8, wherein the mixing unit and the reservoir are reversibly connected or connectable to one another via a first form closure.
10. The device according to claim 8, wherein the closure element is a rotary valve, through which the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit extends and which, in a first rotary valve position, leaves the inlet channel and the outlet channel in the first channel position and, by rotating to a second rotary valve position, moves the inlet channel and the outlet channel to the second channel position.
11. The device according to claim 9, wherein, after the reservoir has been separated from the mixing unit by releasing the first form closure, the closure element is movable into a closure position in order to close, in terms of fluid conduction, the part of the two-part inlet channel and the

two-part outlet channel that faces the mixing unit.
12. The device according to claim 11, wherein the closure element is a screw.
13. The device according to claim 9, wherein the mixing unit and the reservoir are reversibly connected or connectable to each other via a second form closure.
14. A method for providing a bone cement paste from two starting components using the device of claim 1, the method comprising: a. flowing the monomer liquid from the reservoir through the inlet channel into the distal part of the interior space while simultaneously discharging a gas from the interior space through the outlet channel into the reservoir, b. conveying the monomer liquid from the distal part of the interior space through the conduit means into the proximal part of the interior space by means of advancing the conveying piston in the direction of the discharge piston.
15. The method according to claim 14, wherein, prior to conveying the monomer liquid in step b., the part of the two-part inlet channel and the two-part outlet channel that faces the mixing unit is closed in terms of fluid conduction by the closure element.